Graded Assessment - Hash Functions

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1. The set of all inputs that produce the same hash value is known as — **1 / 1 point**

- (x) the preimage.
- ( ) the collision space.
- ( ) the input partition.
- ( ) a collision.

Correct
Correct.

2. The effect of a Birthday Attack is to — **1 / 1 point**

- (x) effectively cut the width of the digest in half when looking for two random messages whose digests collide.
- ( ) find a preimage for a given digest in significantly less time than brute force.
- ( ) significantly improves the performance of a second-preimage attack.
- ( ) reduce the complexity of the attack to the order of the number of bits in the digest.

Correct
The effectiveness of a Birthday Attack is related to the square root of the number of digests, which translates to a digest of half the width.

3. If Alice send Bob a plaintext message along with the digest that has been encrypted with her private key, — **1 / 1 point**

- ( ) then Bob can verify the integrity, but not the authenticity of the message.
- ( ) then Bob can't confirm either the integrity or the authenticity of the message because he would also need her private key to decrypt the digest.
- ( ) then Bob can verify the authenticity, but not the integrity, of the message.
- (x) then Bob can verify both the integrity and the authenticity of the message.

Correct
As long as Alice's private key has not been compromised, she is the only person that could have provided that enciphered digest.

4. The time complexity of a brute force preimage attack against an N-bit hash function scales — **1 / 1 point**

- (x) exponentially with N.
- ( ) exponentially with N/2.
- ( ) linearly with N.
- ( ) as the square root of N.

Correct
Correct.

5. Even though preimage and second preimage attacks against most hash functions are nearly equivalent, they are treated separately because — **1 / 1 point**

- ( ) cryptanalysts are purists.
- ( ) the attacks are still completely different.
- (x) they might not be equivalent.
- ( ) it allows hash function developers to claim more attacks against which their function is resistant.

Correct
That is sufficient to make it a bad idea to assume that they won't be different.

6. The goal of a hash collision attack is to — **1 / 1 point**

- (x) find any two messages that have the same digest.
- ( ) A message that hashes to the same digest as a target message.
- ( ) A message that has a specific target digest.
- ( ) find messages that are the same as their digest.

Correct
The value of the digest is immaterial.

7. The effectiveness of birthday attacks is evidenced by — **1 / 1 point**

- ( ) the width of modern hash functions.
- ( ) the ease with which they can be used to carry out a preimage attack.
- ( ) the fact that they have a been given a specific name.
- (x) the fact that a significant fraction of breaks into real systems involve this type of attack.

Correct
Correct.

8. Strong avalanche behavior is important in order for — **1 / 1 point**

- (x) messages and digests to be uncorrelated.
- ( ) the hash function to be non-invertible.
- ( ) the hash function to be computationally fast.
- ( ) the hash function to be deterministic.

Correct
A single bit change anywhere in the message should flip about half of the bits, randomly distributed, in the digest.

9. Noninvertibility is required for a hash function to be — **1 / 1 point**

- (x) preimage resistant.
- ( ) hash collision resistant.
- ( ) computationally efficient.
- ( ) deterministic.

Correct
If the hash function is invertible, then finding a preimage for a given digest is straightforward.

10. Which of the following is NOT true of an ideal hash function? — **1 / 1 point**

- ( ) The same input always produces the same output.
- (x) No two inputs produce the same output.
- ( ) The length of the input can be any value.
- ( ) Changing any bit in the input causes about half of the bits in the output to change.

Correct
This would require that the length of the output be longer than the longest possible input, which is unbounded.